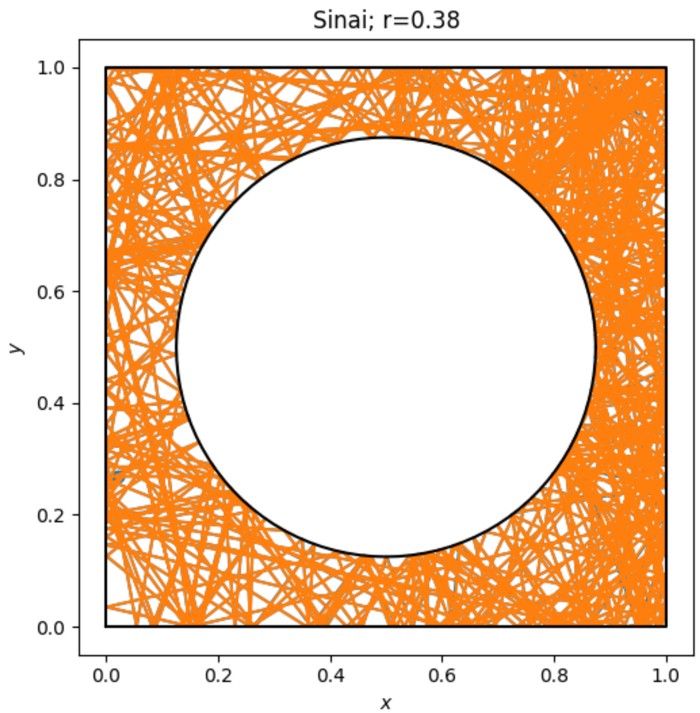
Teorija dinamičnih sistemov



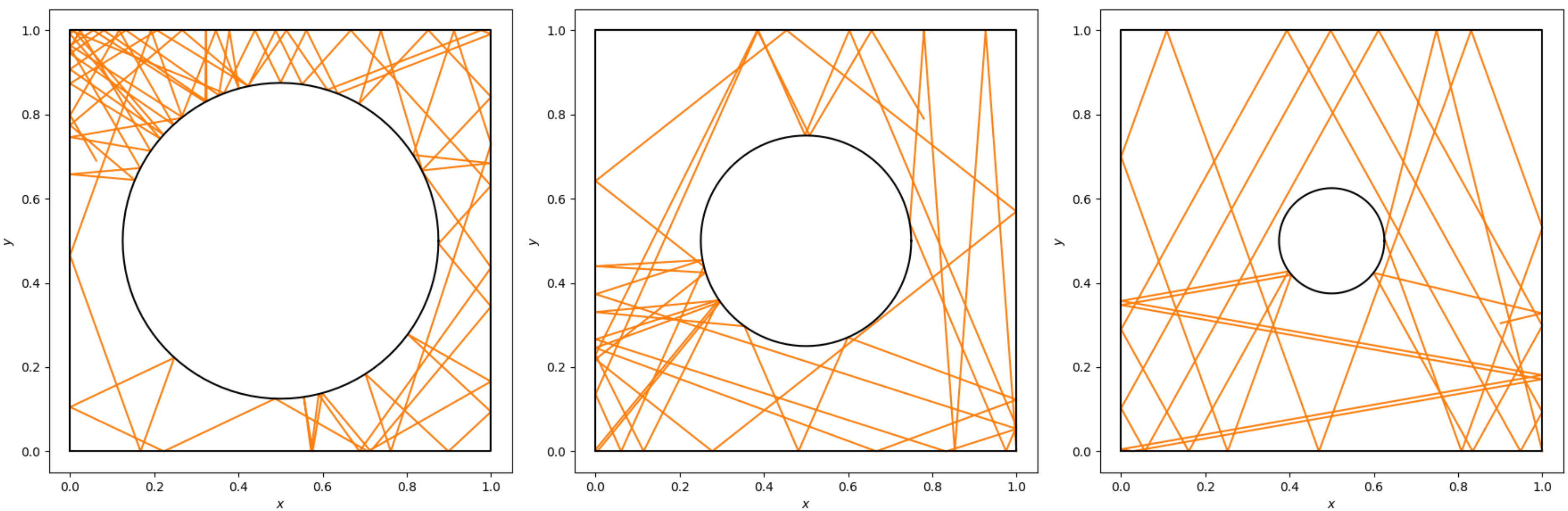
Naloga 25: Sinajev biljard

Žan Butala

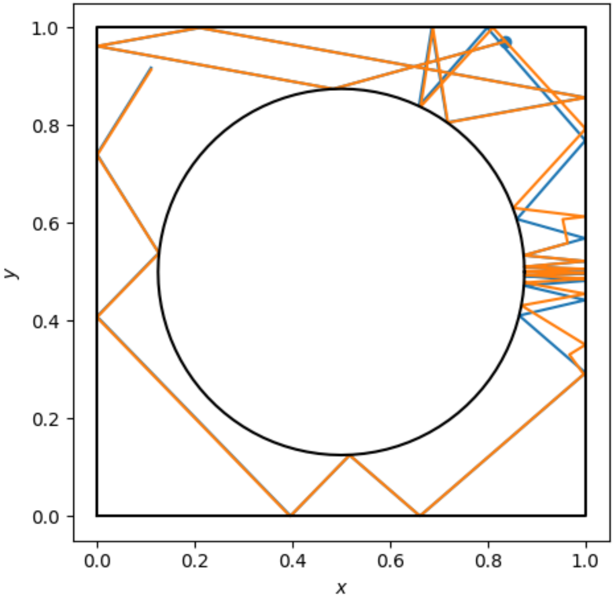
18.1.2023

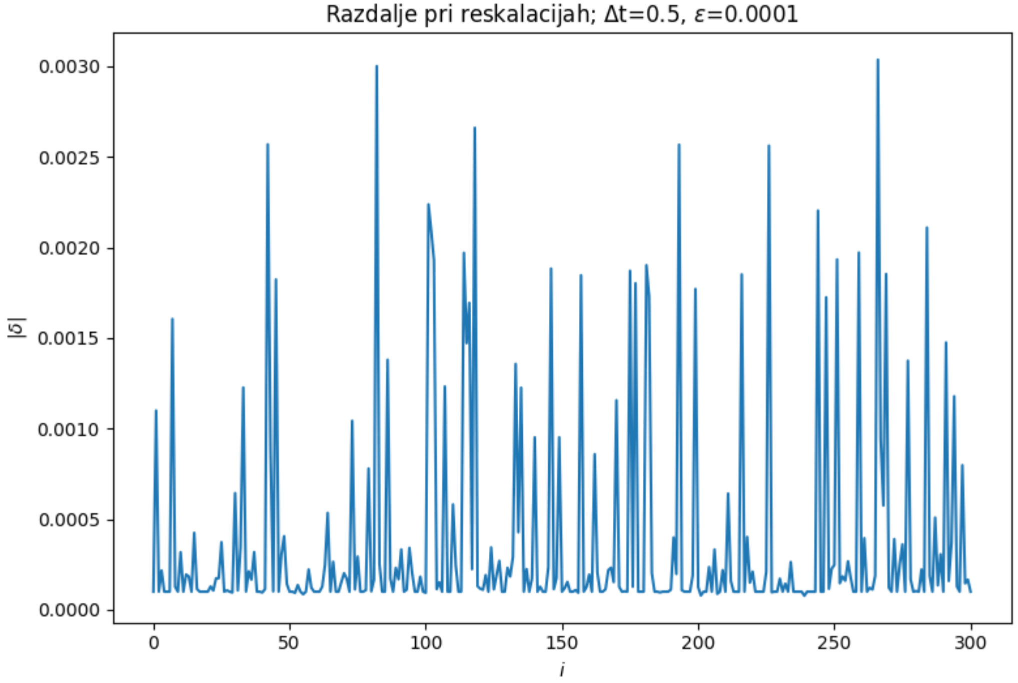
Naloga:

- Izračunati Ljapunove eksponente v odvisnosti od polmera kroga
- Analizirati stabilnost periodičnih orbit

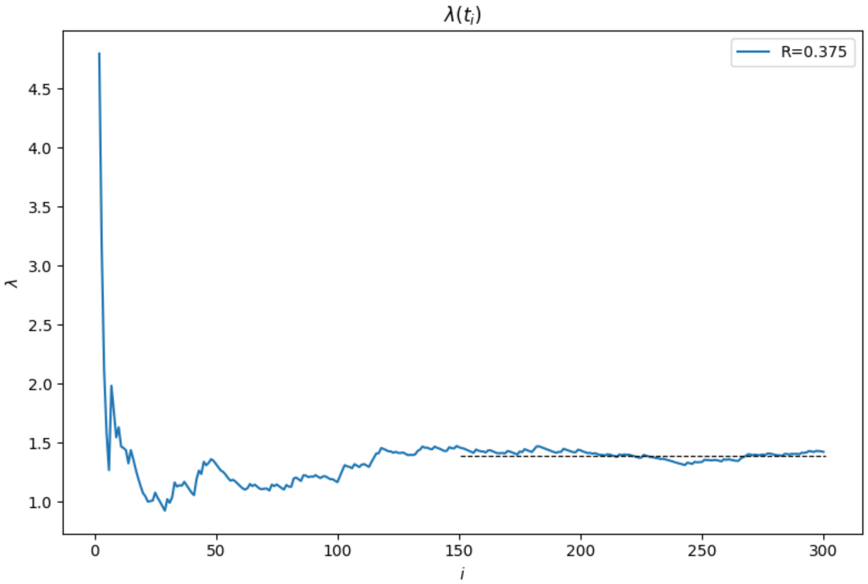


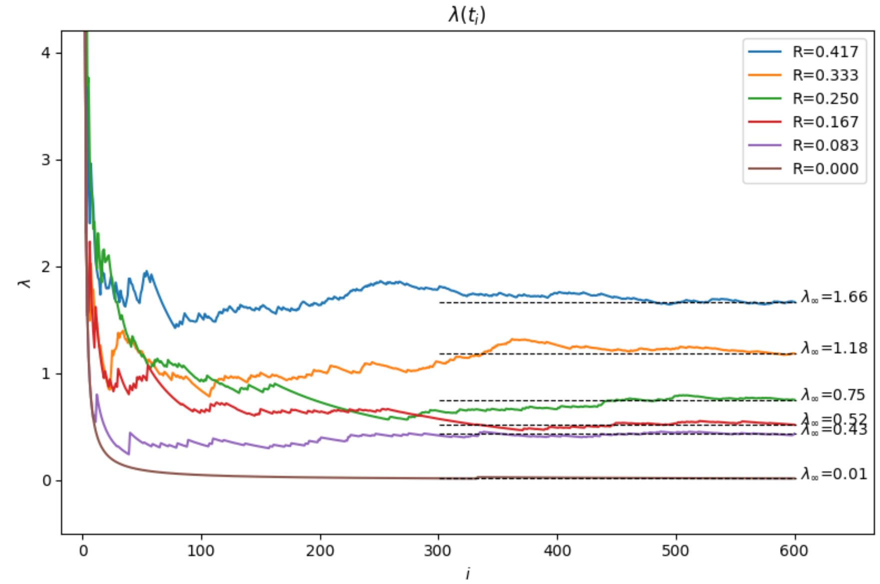
Ljapunov eksponent

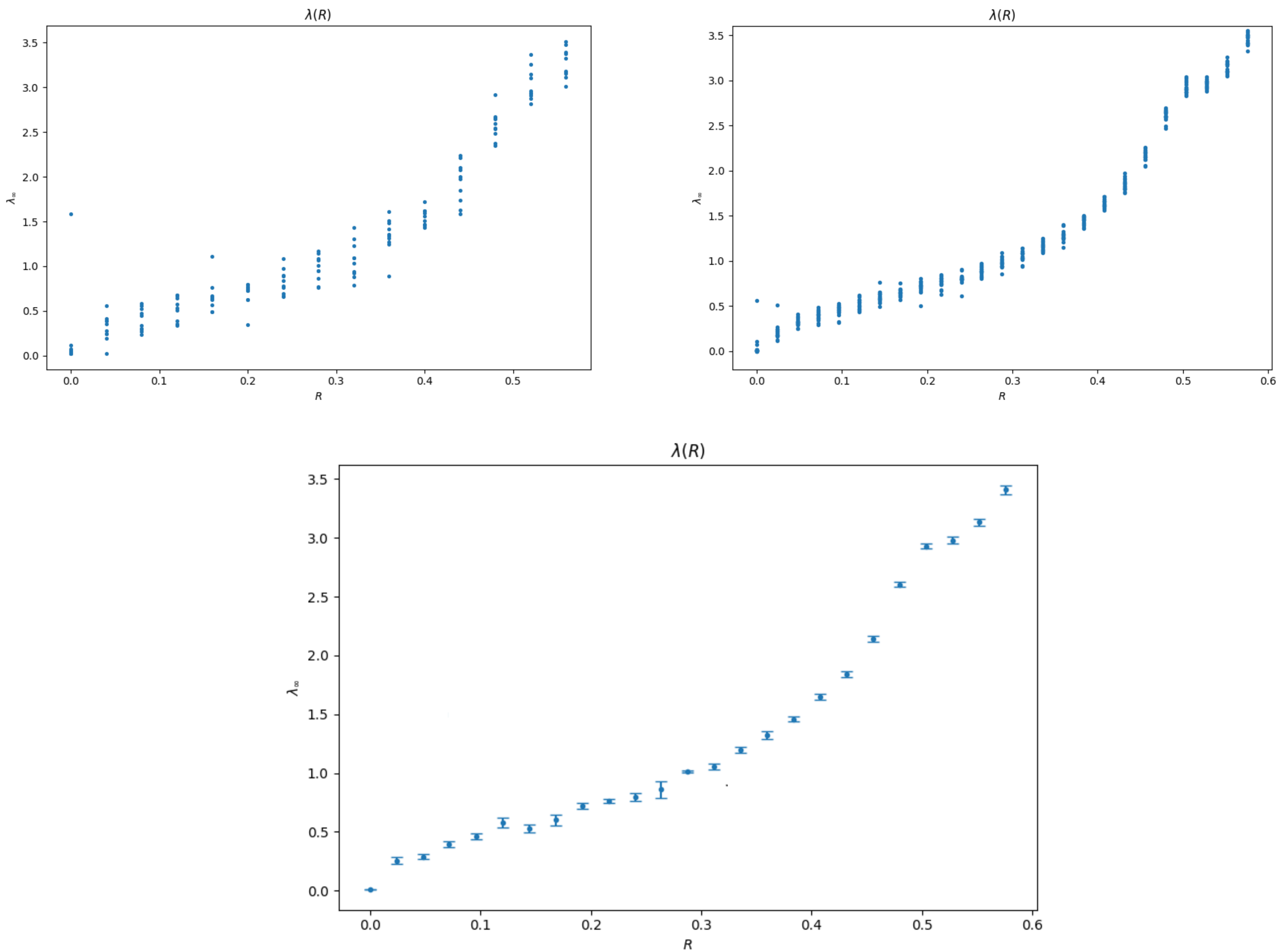




$$\lambda = \lim_{t\to\infty} \frac{1}{t}\left(\sum_i \ln \frac{|\delta(t_i)|}{\varepsilon} \right)$$

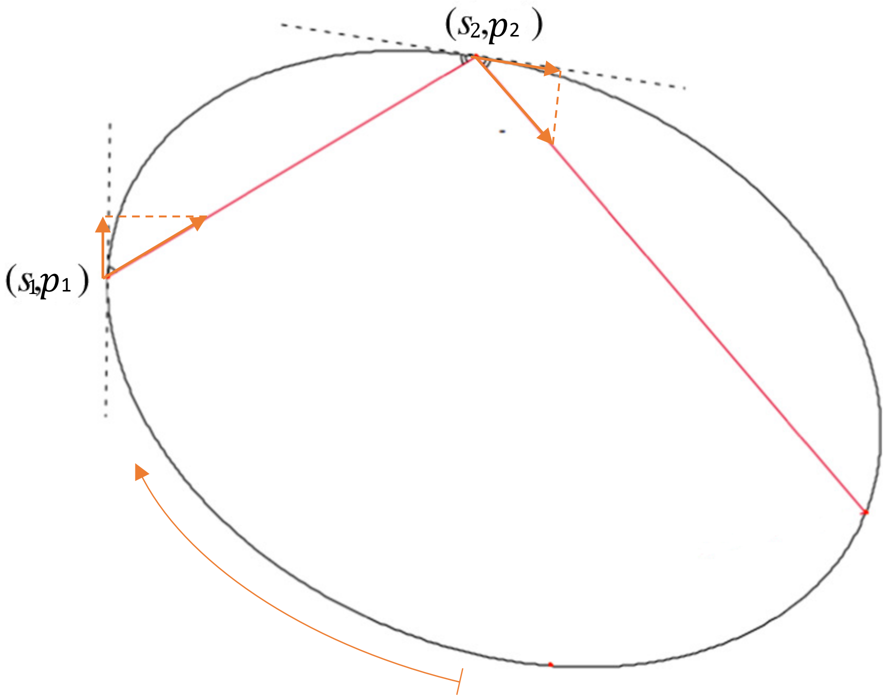


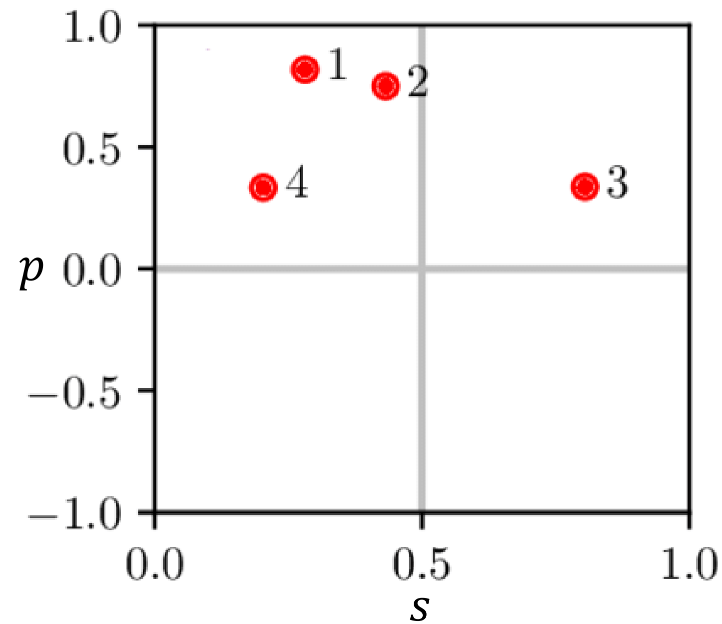




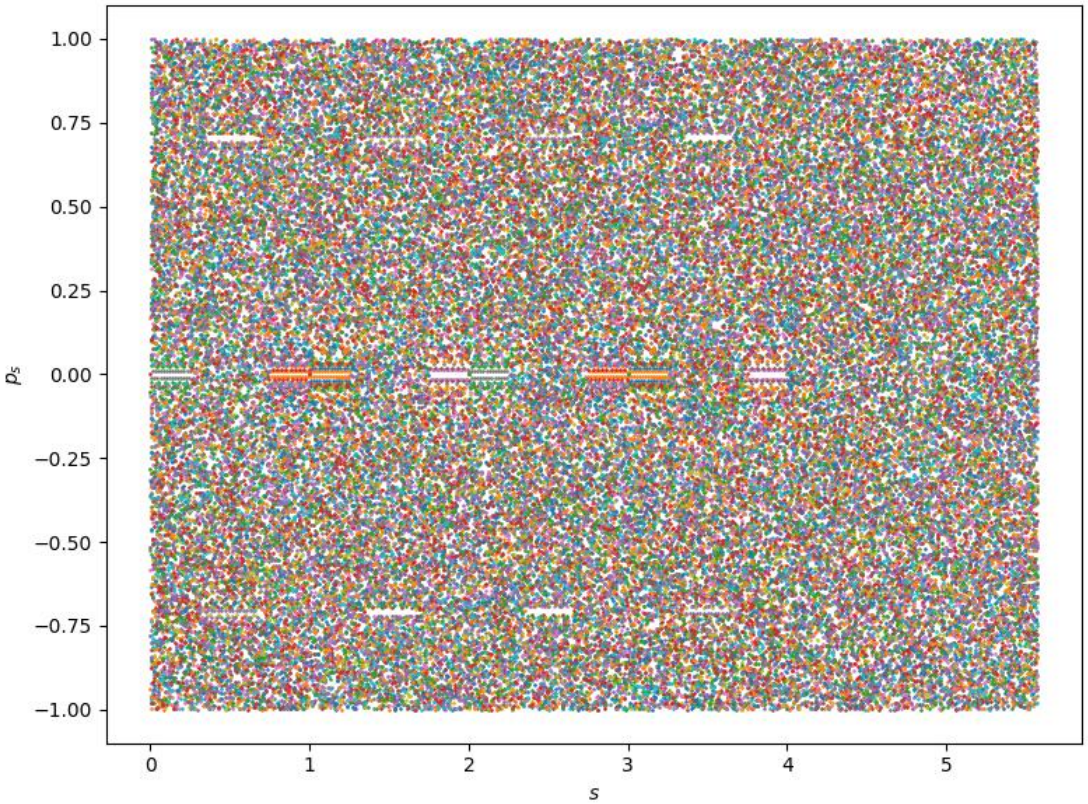
$\lambda(R)$
λ_∞
R
$\lambda(R)$
λ_∞
R
$\lambda(R)$
λ_∞
R

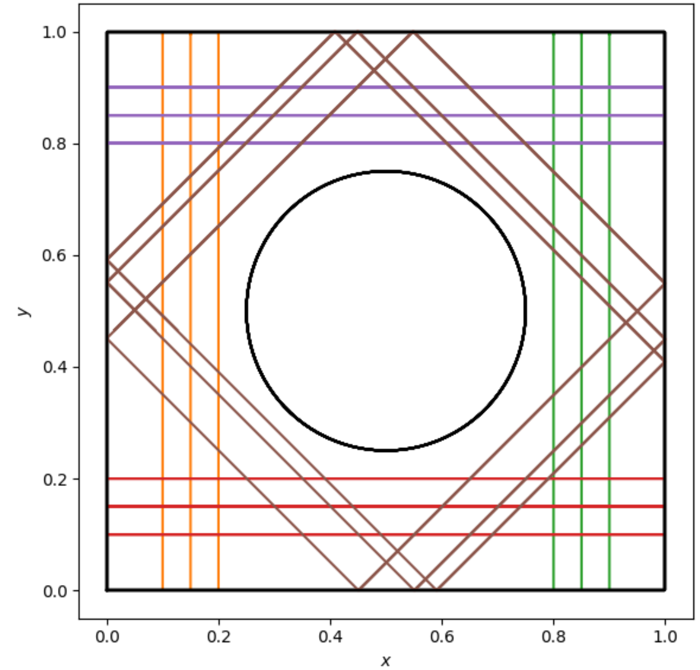
Fazni prostor v P-B koordinatah



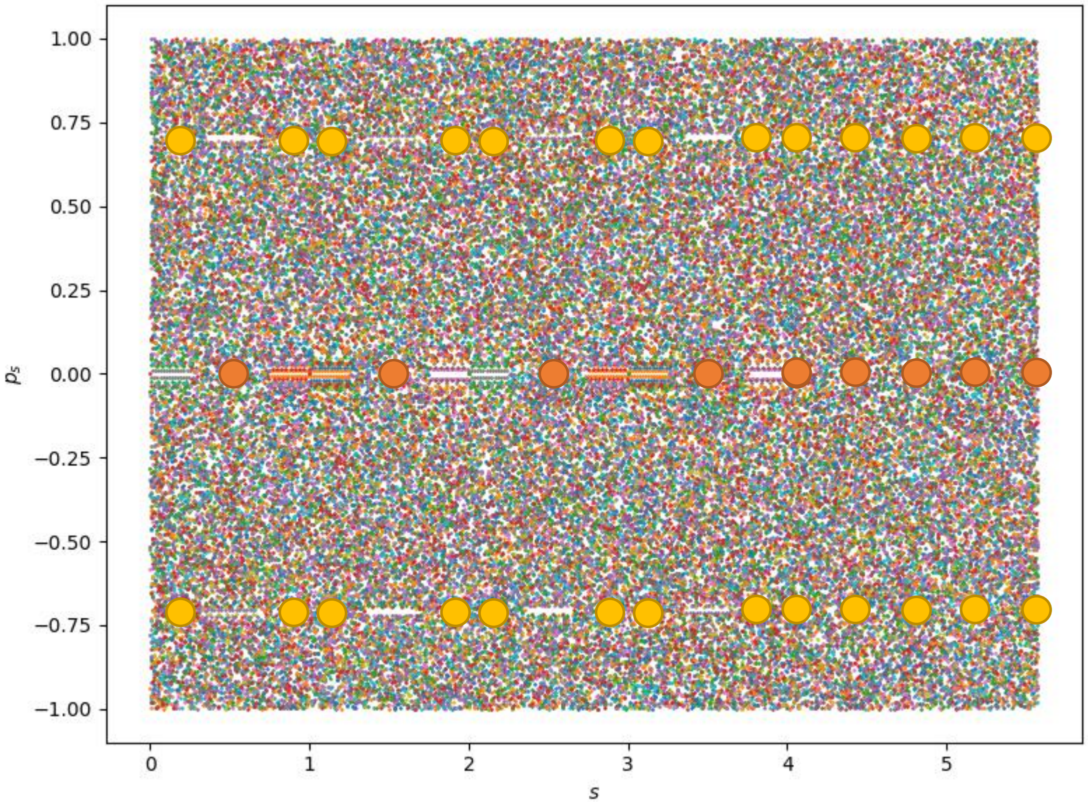


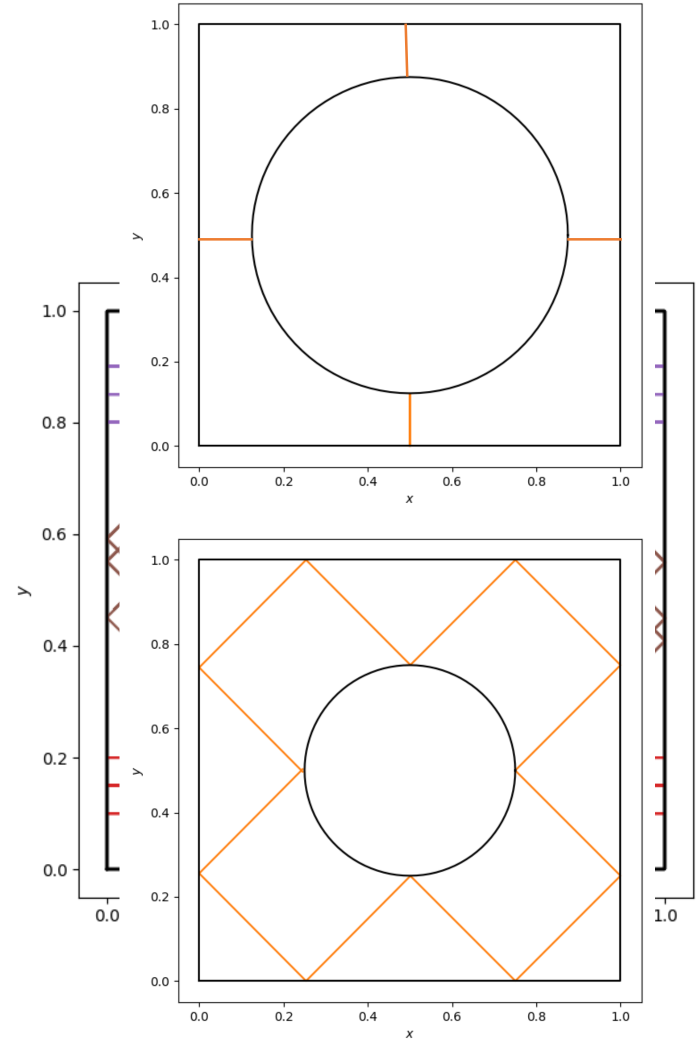
Fazni prostor v P-B koordinatah



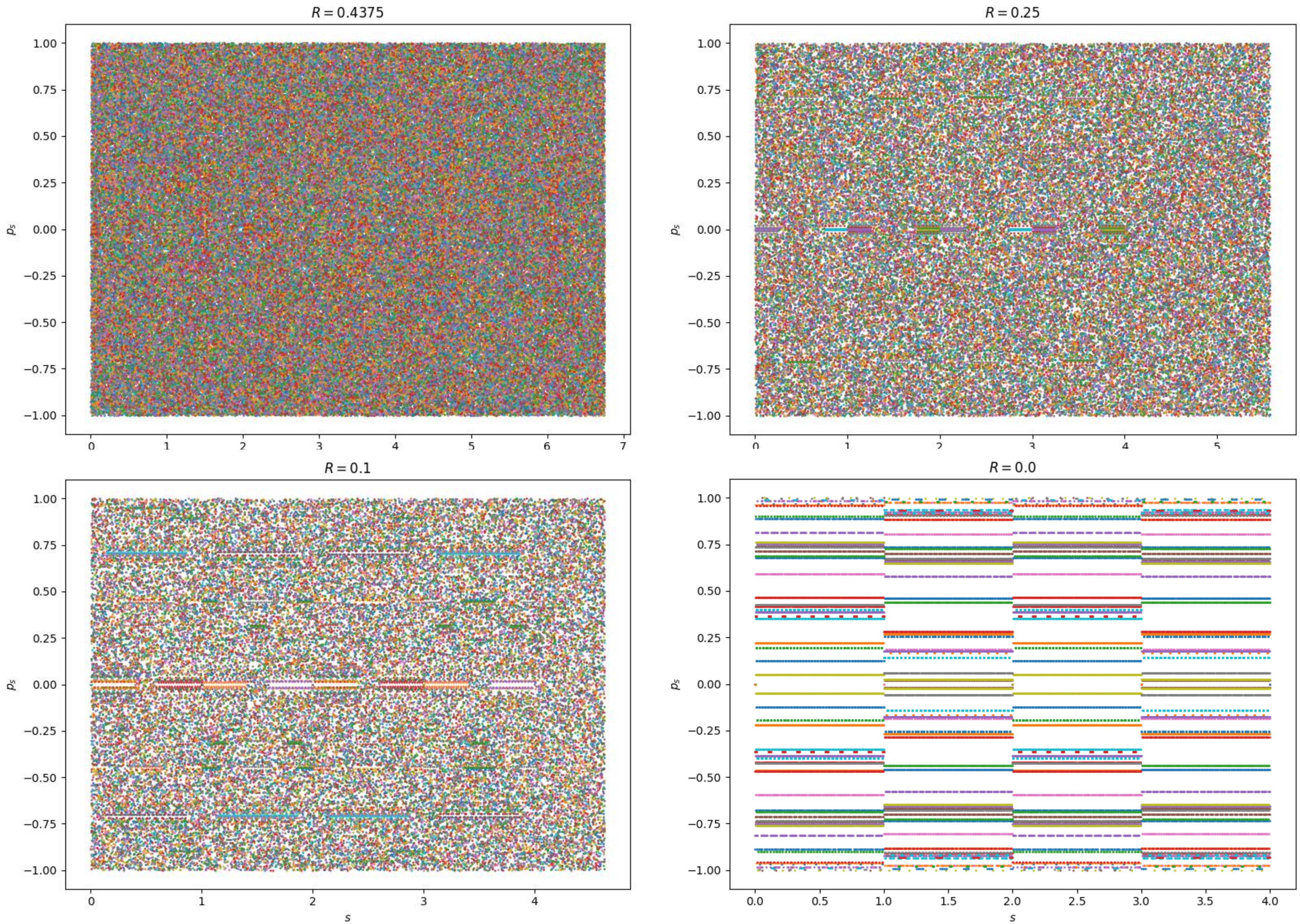


Fazni prostor v P-B koordinatah





Fazni prostor v P-B koordinatah



Stabilnost $(z, \theta \ll 1)$:

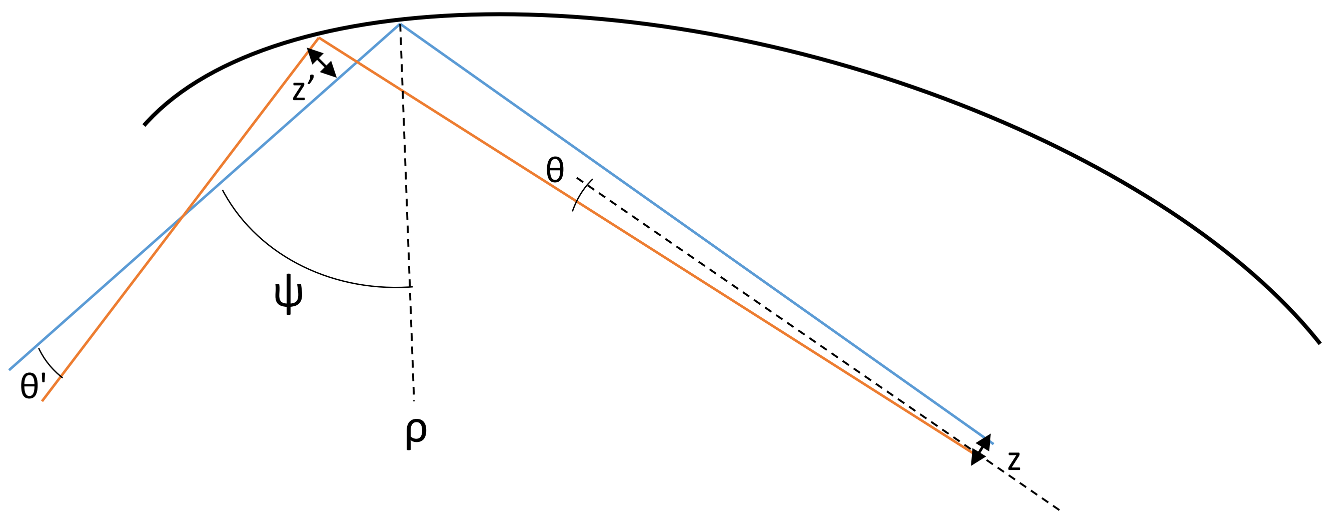
$$\begin{bmatrix} z' \\ \theta' \end{bmatrix} = F_{odboj} F_{prost} \begin{bmatrix} z \\ \theta \end{bmatrix}$$

$$F_{odboj} = \begin{bmatrix} -1 & 0 \\ \frac{2}{\rho \cos \psi} & -1 \end{bmatrix}$$

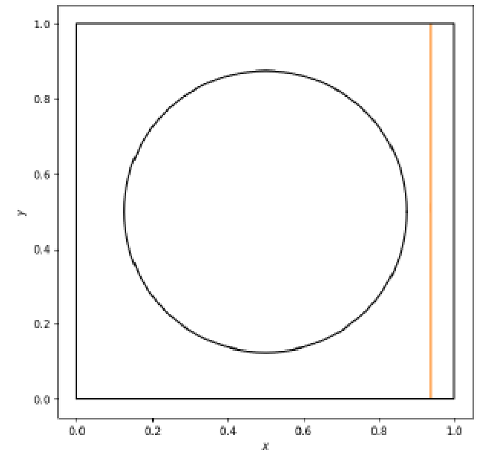
$$F_{prost} = \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix}$$

Pogoj za stabilnost:

$$-2 \leq \mathrm{tr} F \leq 2$$



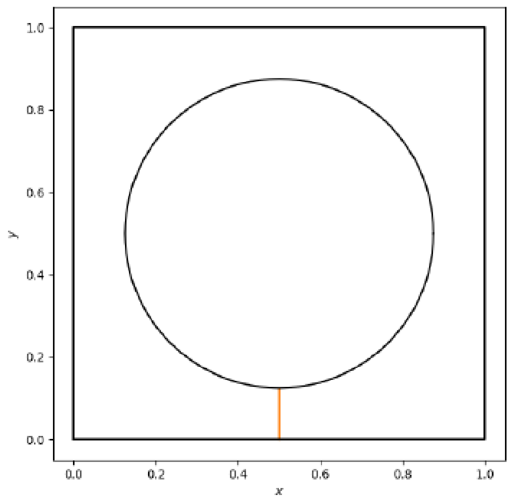
- $\psi_{1,2} = 0,\ \rho_{1,2} = \infty$

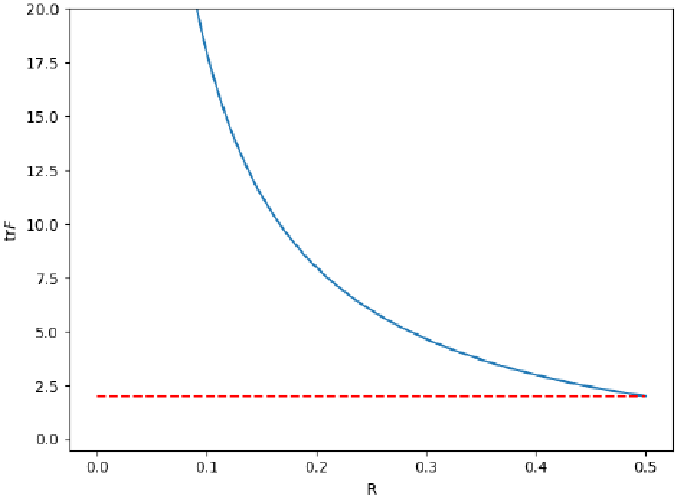


$$\begin{aligned} F =& F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\ =& \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \\ =& \begin{bmatrix} 1 & 2l \\ 0 & 1 \end{bmatrix} \end{aligned}$$

$$\mathrm{tr} F = 2$$

- $\psi = 0,\ \rho_2 = -R$

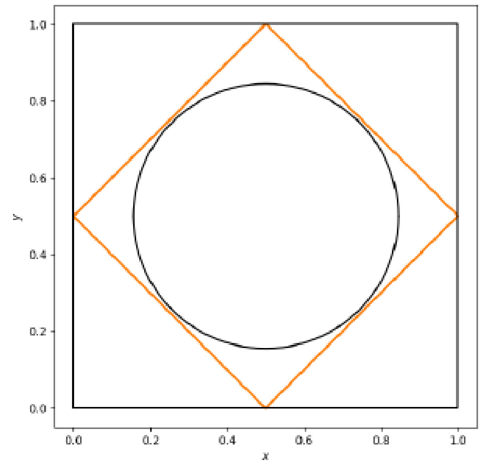




$$
\begin{aligned}
F =& F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\
=& \begin{bmatrix} -1 & 0 \\ \frac{2}{-R} & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \\
=& \begin{bmatrix} 1 & 2l \\ \frac{2}{R} & 1 + \frac{4l}{R} \end{bmatrix} \\
=& \begin{bmatrix} 1 & 1 - 2R \\ \frac{2}{R} & 1 + \frac{2}{R} - 4 \end{bmatrix}
\end{aligned}
$$

$$
\mathrm{tr}F = -2(1 - \frac{1}{R})
$$

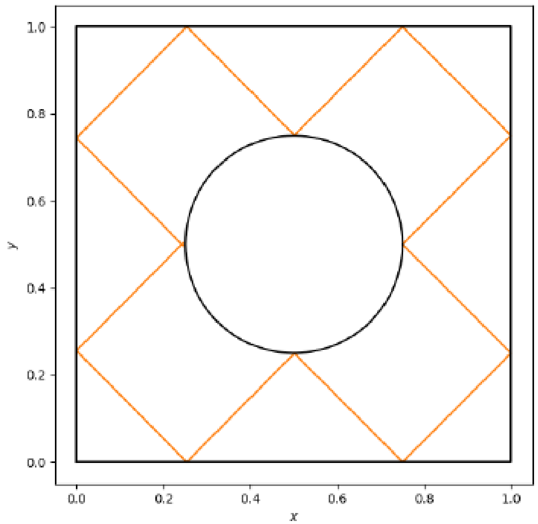
- $\psi_{1,2,3,4} = \frac{\pi}{4}$, $\rho_{1,2,3,4} = \infty$, $R < \frac{\sqrt{2}}{4}$

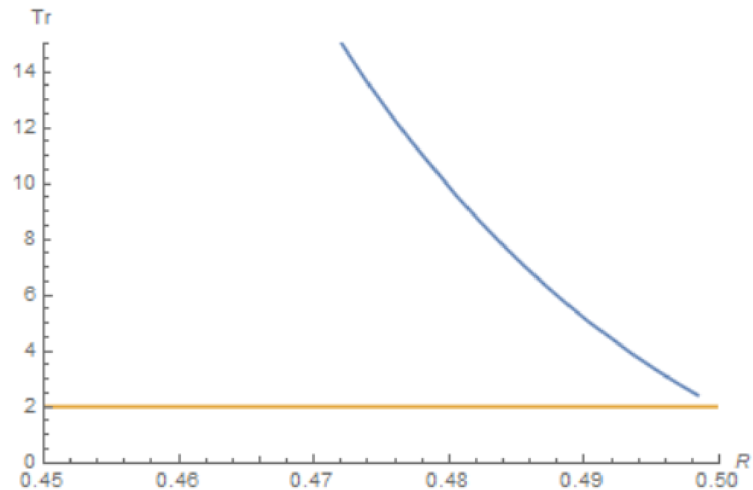


$$
\begin{aligned}
F =& F_{odb_4} F_{prost_4} F_{odb_3} F_{prost_3} F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\
&= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 4l \\ 0 & 1 \end{bmatrix}
\end{aligned}
$$

$$\mathrm{tr} F = 2$$

- $\psi_i = \frac{\pi}{4}$, $\rho \in \{-R, \infty\}$





$$
\begin{aligned}
F =& F_{odb_{12}} F_{prost_{12}} \ldots F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\
=& \ldots \ Mathematica
\end{aligned}
$$

$$\mathrm{tr} F > 2$$

$$l1 = \sqrt{(1/2 - R)^2 + (1/2 - R)^2};$$

$$l2 = \sqrt{(1/2 - (1/2 - R))^2 + (1/2 - (1/2 - R))^2};$$

```
MatrixForm[MatrixPower[Dot[A1, Dot[B1, Dot[A2, Dot[B2, Dot[A2, Dot[B1]]]]]], 4]]
```

$$\begin{pmatrix} \left(1+\frac{2\sqrt{2}\left(-2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}-\sqrt{2}\sqrt{R^2}\right)}{R}\right)^2+\left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}\right)}{R}\right)}{R}\right)\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}+\left(-2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}-\sqrt{2}\sqrt{R^2}\right)\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}\right)}{R}\right)\right) & \left(1+\frac{2\sqrt{2}\left(-2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}-\sqrt{2}\sqrt{R^2}\right)}{R}\right)\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}+\left(-2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}-\cdots\right.\right. \\ \left(1+\frac{2\sqrt{2}\left(-2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}-\sqrt{2}\sqrt{R^2}\right)}{R}\right)\left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}\right)}{R}\right)}{R}\right)+\left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}\right)}{R}\right)}{R}\right)\left(\frac{2\sqrt{2}\left(-2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}-\sqrt{2}\sqrt{R^2}\right)}{R}+\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}\right)}{R}\right)^2\right) & \left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}\right)}{R}\right)}{R}\cdots\right. \end{pmatrix}$$

```
Simplify[Tr[MatrixPower[Dot[A1, Dot[B1, Dot[A2, Dot[B2, Dot[A2, Dot[B1]]]]]], 4]]]
```

$$\frac{2\left(6049 R^4+32 R^2\left(125+32\sqrt{(-1+2R)^2}\right)-32 R^3\left(244+105\sqrt{R^2}+57\sqrt{(-1+2R)^2}\right)-256 R\left(4+\sqrt{(-1+2R)^2}+\sqrt{R^2}\left(3+8\sqrt{(-1+2R)^2}\right)\right)+64\left(2+8\sqrt{R^2}\sqrt{(-1+2R)^2}+3\left(R^2\right)^{3/2}\left(16+15\sqrt{(-1+2R)^2}\right)\right)\right)}{R^4}$$

```
Plot[{%71, 2}, {R, 0, 5}, PlotRange → {{0.45, 0.5}, {0, 15}}, AxesLabel → {HoldForm[R], HoldForm[Tr]}]
```